

Starrydata2 snapshot

22 Aug 2026
9,494 papers
156,101 curves

Cleaning

11 stages
280,664 rows

Featurization

132 MAGPIE
+ 264 CBFV
+ T
= 397 features

Per-target datasets

S 185,064
 σ 182,755
 κ 121,110
zT 129,419

(A) §3.1 Validation ladder

ungrouped:
random 80/20,
5-fold, 10-fold

grouped
(5 × 5 CV):
composition,
chemistry
cluster

+ attribution
control (§3.2):
TreeSHAP,
random vs
grouped

(B) §3.3 Label-noise ceiling

round-robin
measurement
noise

+ teMatDb
digitization
noise

(C) §3.4 Descriptor ablation

feature sets:
MAGPIE,
CBFV,
full

(D) §3.5 External transfer

ESTM,
teMatDb;
in-support
restriction

(E) §3.6 Direct vs derived zT

zT predicted
directly vs
 $S^2\sigma T/\kappa$
from predicted
components

56,088-row
subset

XGBoost, frozen per-target hyperparameters

same model in every model-based analysis